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Does ChatGPT provide better advice than robo-advisors?

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ABSTRACT

We develop three investor profiles with different risk attitudes and consult ChatGPT and 17 robo-advisors to recommend an investment portfolio. We compare the recommendations with a benchmark derived from academic literature. ChatGPT's recommendations align with the three investor profiles and the benchmark. In contrast, only three out of the 17 robo-advisors come close to meeting the benchmark for all three investor profiles. Three robo-advisors fail to meet the benchmark for all investor profiles. Our findings reveal that ChatGPT provides better financial advice for one-time investments than robo-advisors. A policy implication is to clearly disclose that independent sophisticated chatbots can be recommended as a trustworthy source of information for retail investors.

1. Introduction

Identifying and using the most relevant information in financial decisions is something that many retail investors find challenging (Oehler et al., 2023). A significant percentage of retail investors suffer from information overload and insufficient financial literacy to effectively utilize the available information (Stolper and Walter, 2017). As a result, these retail investors often rely on financial advisors (Foerster et al., 2017). Traditionally, financial advisors were human. However, a decade ago, robo-advisors were introduced to the market and have since become a permanent fixture, often as part of larger financial service providers such as banks, fund companies, and insurance companies. Robo-advisors were considered groundbreaking and promising, especially for their ability to provide financial advice with minimal behavioral biases typically associated with human advisors (D'Acunto et al., 2019; Foerster et al., 2017; Inderst and Ottaviani, 2009; Linnainmaa et al., 2020). A robo-advisor typically constructs portfolios tailored to investors' preferences by asking questions about risk preferences, loss tolerance, investment amount, financial reserves, monthly disposable net income, and investment experience. Based on the answers to these questions, robo-advisors recommend an investment portfolio typically consisting of stock and bond index funds (D'Acunto and Rossi, 2020; Horn et al., 2020; Horn and Oehler, 2020; Kaya, 2017; Oehler et al., 2022; Rossi and Utkus, 2020).

Technological progress is opening new possibilities for financial advice. Companies in the financial service sector are expected to allocate more than half of their planned investment budget on artificial intelligence (AI) in 2023 and 2024, with a focus on improving

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customer-facing processes using chatbots (Sarathy, 2023). Currently, the most discussed chatbot is ChatGPT. It was introduced in November 2020 and is a conversational variant of the third iteration of the Generative Pre-Trained Transformer (GPT) model (Brown et al., 2020). Kim et al. (2023) describe the model as pre-trained on a vast language corpus and then fine-tuned for specific tasks, such as summarization or sentiment analysis. GPT outperforms other existing models in summarization tasks (Bhaskar et al., 2022; Goyal et al., 2022) and performs impressively on financial literacy tests (Niszczota and Abbas, 2023). Therefore, it appears straightforward to analyze ChatGPT's ability as financial advisor. This has not been done yet although ChatGPT's utility in finance research and practical finance tasks have been the subject of other studies (e.g. Dowling and Lucey 2023, Kim 2023). We address this gap in the literature by analyzing the research question whether ChatGPT provides better advice than robo-advisors.

To answer this research question, we develop three investor profiles primarily differing in their risk attitudes. In April 2023, we utilized ChatGPT and 17 globally available robo-advisors to derive recommendations for each investor profile. We compare these recommendations with a benchmark from academic literature regarding a suitable portfolio structure.

Our findings reveal that ChatGPT provides better financial advice for one-time investments than robo-advisors. While only three out of the 17 robo-advisors come close to meeting the benchmark for all three investor profiles, three robo-advisors failed to meet the benchmark for any investor profile. ChatGPT, on the other hand, provides recommendations and additional information that aligned with the three investor profiles and the benchmark from academic literature. This is remarkable since ChatGPT used only a few targeted questions, and the consultation took significantly less time than answering the questions posed by the robo-advisors. Our findings are relevant not only to academics but also to regulators, investors, and providers of financial services.

We contribute to the literature on financial advice, AI for financial services, and financial information for retail investors in four ways. First, we derive three investor profiles from the academic and practitioner literature that can be used to analyze the suitability of financial advice. Second, we demonstrate the economic usefulness of generative AI-based techniques for analyzing and summarizing investment information for different investor profiles in a meaningful way. Third, we provide preliminary evidence that chatbots can significantly improve the information provided to investors by enhancing presentation and content. ChatGPT retains and amplifies information content, allowing investors to focus on information that truly matters. Fourth, as investors often face significant information processing costs or investment commissions, the generation of investment approaches *via* ChatGPT could support investors' decision-making and increase investment returns.

Our findings imply that robo-advisors and even human advisors could use the information provided by ChatGPT to streamline their advice process and enhance the quality of advice. This finding can also be utilized by policies regarding retail investor information documents. Independent chatbots can be recommended as a trustworthy source for a second opinion to double-check recommendations provided by a human or robo-advisor. In summary, our findings indicate that generative AI systems have the potential to be the next meaningful advance in financial advice and investor information, which could be beneficial for service providers (lower costs) and investors (better advice).

Our study is organized as follows: In the next section, we review the related literature. In Section 3, we introduce the data and methodology to generate portfolio recommendations with ChatGPT and robo-advisors. Section 4 presents the empirical results. We discuss our findings and conclude in Section 5.

2. Related works

2.1. ChatGPT and robo-advisors

ChatGPT employs AI to provide easily accessible information and has even been utilized in writing books about the usage of machine learning in finance, e.g., to develop trading strategies (Gautier, 2022; Osterrieder and ChatGPT, 2023). Furthermore, ChatGPT shares some advantages with robo-advisors, such as 24/7 availability and low consultation costs. These factors make both services intriguing to individuals with low net wealth who cannot afford a human advisor (D'Acunto and Rossi, 2020; Horn et al., 2020; Rossi and Utkus, 2020).

In general, particularly younger investors do not exhibit algorithm aversion and are willing to follow algorithm-based financial advice (Germann and Merkle, 2022; Isaia and Oggero, 2022). Li et al. (2022) discovered that investors invest more when the portfolio is constructed by a robo-advisor than when a human-advisor is involved. The reason behind this is that investors trust the robo-advisor more than human advisors, even though both types of advisors are a source of uncertainty. The results align with the theory of source preference (Chew and Sagi, 2008; Fox and Tversky, 1995) and findings in Holzmeister et al. (2023). According to these findings, robo-advisors provide a simpler path to initial or one-time financial investments than human advisors, because the "source" of the advice is more trusted.

Although the discussion about the potential replacement of human advisors by robo-advisors has only just begun (Back et al., 2023; D'Acunto et al., 2019; Holzmeister et al., 2023; Li et al., 2022; Oehler et al., 2022), it is clear that chatbots like ChatGPT can be the next technological innovation to better cater to retail investors' needs and preferences. However, it remains uncertain whether retail investors could benefit from ChatGPT as financial advisor. In other fields, such as medicine or safety-related advice, ChatGPT sometimes provides misinformation that has the potential to harm users (Oviedo-Trespalacios et al., 2023; Kleesiek et al., 2023).

2.2. Financial advice to retail investors

The literature on financial advice for retail investors links the recommended asset allocation to investors' risk attitude (e.g. Choi, 2022; Foerster et al., 2017). Foerster et al. (2017) find that the least risk-tolerant allocate, on average, 40 % of their portfolio to risky assets, such as stocks or equity funds. In contrast, the most risk-tolerant investors invest 80 % in risky assets. Jacobs et al. (2014) conduct a literature review and identify a consensus recommendation of approximately 60 % in stocks and 40 % in bonds as a one-size-fits-all asset allocation. Some sources suggest investing up to 10 percentage points of the equity share in alternative assets like commodities (e.g. Jacobs et al. 2014, and the studies cited therein). These asset allocations have historically provided high risk adjusted returns and, as such, are the best estimate for efficient portfolios in the future. However, it remains unclear whether robo-advisors and ChatGPT recommend similar portfolios based on investors' risk attitudes.

Previous studies have shown that human financial advisors often recommend high-cost assets, such as actively managed mutual funds, to maximize their own sales (Stolper and Walter, 2017). This practice is detrimental to retail investors as the fees reduce their returns (Fama and French, 2010; French, 2008). While some robo-advisors recommend low-fee funds, others exclusively promote the rather expensive funds of few companies. Furthermore, the rebalancing service provided by robo-advisors can be costly and may not be beneficial for retail investors (Horn and Oehler, 2020). Since the internet, which serves as ChatGPT's sole information source, is filled with marketing material for high-fee funds and rebalancing services, we are particularly interested in whether ChatGPT can help avoid such costly mistakes. To the best of our knowledge, this question has not been analyzed thus far, even though it is highly relevant for the financial performance of retail investors.

3. Data and methodology

3.1. Three investor profiles

In accordance with the literature (e.g. Oehler et al. 2019, Choi 2022, Foerster et al. 2017, Jacobs et al. 2014) and tests and reviews of robo-advisors on websites and consumer protection organizations, we establish three investor profiles primarily differing in their risk attitude. These profiles need to be suitable for answering the questions posed by the robo-advisors and must have clear specifications (personal preferences) for ChatGPT. Simultaneously, the investor profiles should represent real-world investors and be selected in a manner that aligns with the guidance found in academic literature regarding a recommended portfolio structure, which serves as a benchmark for actual investment decisions and financial advice. The following three investor profiles are distinct:

Investor profile A: High risk aversion, with a loss tolerance of approximately five percent.
Investor profile B: Moderate risk aversion, with a loss tolerance of around ten percent.
Investor profile C: Low risk aversion, with a loss tolerance of approximately twenty-five percent.

For all three investor profiles, we apply an initial and one-time investment amount of 10,000 Euros, an investment horizon of roughly ten years, good investment experience, a three-month liquidity reserve, and a net disposable income of 1,500 Euros (monthly). It's worth noting that some robo-advisors might not recommend investments in risky assets if we opt for a lower liquidity reserve or monthly income.

The most popular robo-advisors typically suggest portfolios consisting of equity and bond index funds. Following the literature (Choi, 2022; Foerster et al., 2017; Jacobs et al., 2014) we anticipate that investor profile A receives the advice to build a portfolio consisting of 60 to 70 % broadly diversified global bond ETFs (and/or fixed-term deposits, short-maturity and top-rated bonds, cash; for better readability we just refer to "bond ETFs") and 30 to 40 % broadly diversified global equity ETFs. Investor profile B's portfolio should be composed of 40 to 50% bond ETFs and 50 to 60 % equity ETFs. For investor profile C, we expect a portfolio constructed of 10 to 20 % bond ETFs and 80 to 90 % equity ETFs. It would be suitable to allocate up to 10 percentage points of the equity share to alternative assets like commodities.

3.2. Generating investment information with ChatGPT

ChatGPT and the robo-advisors were contacted through a private email account to ensure that no conclusions could be drawn regarding a university or a research project. To obtain advice from ChatGPT, we initiate the process with a question that encompasses all the information related to the corresponding investor profile. As ChatGPT is initially cautious in providing clear advice, we must follow up with additional questions to receive a specific portfolio recommendation. We took care not to provide any indications regarding the potential age, gender, or other personal information of the investor, nor any information that would indicate whether we were seeking advice for ourselves or on behalf of others.

For investor profile C, for example, we entered the following three statements by and by: (1) Best way to invest 10,000 Euros for 10 years, disposable net income 1,500 Euros per month, financial reserves 4,500 Euros, risk prone, loss tolerance 25 %, investment experience good?; (2) What portfolio should then form exactly for 10,000 Euros?; (3) Personal goals, preferences and risk appetite are known: high loss tolerance of 25 %, investment for 10 years. After these three inputs, ChatGPT provides a concrete recommendation.

3.3. Constructing portfolios with robo-advisors

To compile a selection of the most widely used and thoroughly tested robo-advisors in worldwide, we relied on two distinct sources. Firstly, the Forbes list "Top-10 Robo-Advisors BY Assets Under Management" offers valuable insights into the market.[2] Secondly, "Best Robo-Advisor: Comparisons and Tests" presented a more comprehensive list of robo-advisors from extensive testing.[3] In total, we considered 31 robo-advisors. We excluded robo-advisors that require a very high minimum investment amount (>20,000 Euro; $N = 3$; Estably, Liqid, Solidvest), those that did not offer concrete recommendations ($N = 3$; My-si, Smavesto, Openbank), those unable to accomodate investor profiles ($N = 2$; SigFig, Evergreen), and those demanding specific information, such as the U.S. social security number ($N = 6$; Vanguard Robo-Advisors, Betterment/Wealthfront, Empower, Acorns, M1, Oskar).

The 17 remaining robo-advisors, listed in alphabetical order, are bevestor, comdirect, Deutsche Bank/Robin, easyfolio, Fidelity, fintego, FutureAdvisor, Gerd Kommer Capital, ginmon, growney, Quirion, Raisin, Scalable Capital, Schwab, Vanguard Invest, Visualvest, Whitebox. Across our three investor profiles, we answered the corresponding questions of the robo-advisors[4], which varied significantly in number and detail, until a portfolio recommendation was provided. However, it is important to note that although robo-advisors might ask many questions to arrive at a recommendation, the underlying investment task is the same as for ChatGPT (an initial one-time investment of 10,000 Euros with a 10-year investment horizon). Hence, we would expect similar recommendations.

4. Results

4.1. ChatGPT

The investment recommendations from ChatGPT for the three investor profiles are clear and align with the benchmark from the academic literature.

For investor profile A (high risk aversion), the portfolio recommendation is as follows: "60 % in a bond fund (this would provide you with a reliable source of income while preserving your capital. Consider investing in a bond fund that holds high-quality bonds with low credit risk), 30 % in an index fund (this would provide you with exposure to a diversified portfolio of stocks with relatively low risk. Consider investing in a low-cost index fund that tracks the performance of a broad-based stock index, such as the S&P 500 or the MSCI World Index), and 10 % in a REIT (this would provide you with exposure to a diversified portfolio of income-producing properties, such as apartments, shopping centers, and office buildings. Consider investing in a publicly traded REIT or a real estate mutual fund)."

For investor profile B (moderate risk aversion), ChatGPT recommends a portfolio with the following allocation: "50 % in a diversified equity index fund (this investment provides exposure to a broad range of stocks, such as those in the S&P 500 or the MSCI World Index. Equity funds can offer higher potential returns than bonds, but also come with higher risk. A low-cost, passive index fund would be suitable for this purpose), 40 % in a diversified bond index fund (this investment provides exposure to a range of bonds, such as U.S. Treasuries or investment-grade corporate bonds. Bonds generally offer lower returns than stocks but can provide a stable source of income and diversify the portfolio), and 10 % in a diversified REIT index fund (this investment provides exposure to real estate investment trusts (REITs), which own and operate income-producing properties. REITs can offer a source of diversification and potentially higher returns than bonds, but also come with higher risk)."

For investor profile C (low risk aversion) with more risk appetite, ChatGPT recommends a portfolio with the following allocation: "70 % in equity mutual funds (this could be divided into different types of funds, such as large-cap, mid-cap, and small-cap funds, as well as funds focused on different sectors or regions. This allocation can provide exposure to potential high-growth companies, but also carries a higher risk of volatility and potential losses), 20 % in bond mutual funds (this could be divided into different types of bond funds, such as government bonds, corporate bonds, and high-yield bonds, to potentially provide some stability to the portfolio), and 10 % in alternative investments (this could include investments such as REITs, commodities, or hedge funds, which can help to diversify

[2] See www.forbes.com/advisor/investing/top-robo-advisors-by-aum/

[3] See https://extraetf.com/de/robo-advisor

[4] Typical questions of robo-advisors are (e.g., growney, selection): (1) The risk of losing some of my money weighs heavily on my mind (strongly disagree, partially agree, mostly agree, strongly agree) (2) The security of an investment is most important to me (strongly disagree, partially agree, mostly agree, strongly agree) (3) I am reluctant to take risks in financial matters (strongly disagree, partly agree, mostly agree, strongly agree) (4) Even small losses make me nervous (strongly disagree, partially agree, mostly agree, strongly agree) (5) I would like to achieve higher returns and am willing to accept risks to do so (strongly disagree, partially agree, mostly agree, strongly agree) (6) The value of financial investments can change up or down every year. Acceptable for me are (−5 % to + 5 %, −10 % to + 10 %, −15 % to + 15 %, below −15 % and above +15 %).Another example is (Fidelity, selection): (1) Of course, you invest this money to make a profit. But what potential loss could you bear if the portfolio were to lose value? (I cannot bear a loss, I could bear a loss of maximum 14 %, 15 %, 16 %, 18 %, 21 %, I could bear a complete loss), (2) I would go for the highest return even if there is a risk involved (Strongly agree, Strongly agree, I don't have a firm opinion on this yet, Somewhat agree, Strongly disagree), (3) How would you describe your attitude when making important financial decisions? (Very Risk Averse, Fairly Risk Averse, Balanced, Fairly Cautious, Very Cautious), What has been your risk tolerance for financial decisions in the past? (Very high, High, Average, Low, Very low), (3) To achieve my financial goal, I prefer an investment that grows slowly but steadily, even if this means a lower return overall (Strongly agree, Strongly agree, I do not yet have a firm opinion on this, Somewhat agree, Not at all agree), (4) In risky situations, I fear, (5) In order to achieve a high return on this investment, you take higher risks (Strongly agree, Strongly agree, I do not yet have a firm opinion on this, Somewhat agree, Not at all agree).

the portfolio and potentially provide higher returns)."

4.2. Robo-Advisors

The 17 robo-advisors present a rather heterogeneous picture. Table 1 provides an overview of the three recommendations from each robo-advisor. For investor profile A, the mean recommendation of 15 robo-advisors is to allocate 66 % to bonds, 31 % to equities, and three percent to other assets like commodities or real estate. However, some robo-advisors deviate significantly; for instance, by recommending a risky portfolio share (equity plus other assets) of more than 60 % (Quirion and Visualvest) or at least 80 % in bonds (fintego, Gerd Kommer Capital, ginmon, Scalable Capital, and Vanguard Invest). Two robo-advisors (Deutsche Bank/Robin and Fidelity) propose no investment due to what they perceive as the investor profile's excessive risk aversion. As a result, nine out of the 17 robo-advisors deviate significantly from the benchmark. This can have significant negative effects on investors, either because they assume too much risk relative to their risk attitude or because they are unlikely to capture the equity premium effectively.

The mean portfolio recommendation for investor profile B is 47 % in bonds, 47 % in equities, and 6 % in other assets. This average recommendation aligns with the benchmark. However, eight robo-advisors significantly miss the benchmark. Comdirect, Quirion, and Visualvest recommend portfolios that are too risky. Deutsche Bank/Robin, Gerd Kommer Capital, Scalable Capital, Schwab, and Vanguard Invest suggest too high a percentage of bonds.

For investor profile C, only four robo-advisors (Gerd Kommer Capital, Quirion, Scalable Capital, and Visualvest) are in line with the benchmark, The remaining robo-advisors recommend either too high or too low risky asset shares. Nevertheless, the average recommendation of all robo-advisors (17 % in bonds, 77 % in equity, 6 % in other assets) aligns with the benchmark.

4.3. Comparison

ChatGPT's portfolio recommendation closely resembles the average recommendation provided by the robo-advisors and the benchmark recommendation derived from academic literature. While we cannot ascertain the exact methodology ChatGPT employs to generate its recommendation, it appears that one of ChatGPT's AI strengths lies in its ability to gather information from a vast number of online sources. As a result, ChatGPT's recommendation may closely align with the consensus of all available sources, representing a market consensus of recommendations, so to speak.

In contrast, robo-advisors rely on their own proprietary models and algorithms to formulate recommendations. Among them, only three robo-advisors come close to meeting the benchmark for all three investor profiles (FutureAdvisor, growney, Raisin), while three robo-advisors fall short of the benchmark for all investor profiles (Deutsche Bank/Robin, Fidelity, Vanguard Invest).

Additionally, ChatGPT generates its recommendations and additional investment information with greater ease and speed, requiring less user data. The supplementary investment information provided by ChatGPT, not fully reported here as it was in the responses to the initial and second input, is well-structured and offers insights into investment categories, checklists, and tips on

Table 1
Recommendations of the 17 robo-advisors, ChatGPT, and academic literature for three investor profiles.

	Portfolio recommendations: Percentage of portfolio invested in Bonds or Bond-ETFs or similar/Equities or Equity-ETFs/Other assets		
Robo-advisor	Typ A (high risk aversion)	Typ B (mod. risk aversion)	Typ C (low risk aversion)
bevestor	70/15/15	48/45/7	48/45/7
comdirect	60/30/10	15/60/25	-/65/35
Deutsche Bank/Robin	–	79/19/2	44/51/5
easyfolio	50/50/-	50/50/-	30/70/-
Fidelity	–	44/23/33	-/90/10
fintego	90/10/-	50/45/5	-/90/10
FutureAdvisor	58/42/-	40/60/-	23/77/-
Gerd Kommer Capital	80/20/-	60/40/-	10/90/-
ginmon	80/17/3	37/55/8	0/87/13
growney	70/30/-	50/50/-	30/70/-
Quirion	30/70/-	20/80/-	10/90/-
Raisin	70/30/-	50/50/-	30/70/-
Scalable Capital	80/20/-	60/40/-	10/90/-
Schwab	60/40/-	60/40/-	34/64/2
Vanguard Invest	80/20/-	60/40/-	0/100/-
Visualvest	39/45/16	29/56/15	20/68/12
Whitebox	75/25/-	50/50/-	1/99/-
Mean robo-advisors	66/31/3	47/47/6	17/77/6
Min-Max robo-advisors	[30–90]/[10–70]/[0–16]	[15–79]/[19–80]/[0–33]	[0–48]/[45–100]/[0–35]
ChatGPT	60/30/10	40/50/10	20/70/10
Benchmark derived from academic literature	60–70/30–40/0–10	40–50/50–60/0–10	10–20/70–80/0–10

Notes: Table 1 displays the recommendations of the 17 robo-advisors for the three investor profiles, including the mean, minimum, and maximum recommended portfolio allocations for bonds, equities, and other assets (e.g. real estate and commodities) for each investor profile. It also includes the recommendation from ChatGPT and the benchmark portfolio recommendation derived from academic literature. Example: The recommendation by the robo-advisor "bevestor" for investor profile A is a portfolio consisting of 70 % in bonds, 15 % in equities, and 15 % in other assets.

investment strategies, diversification, risk-return relationships, and low-cost index funds. In contrast, several robo-advisors predominantly promote their own products with minimal information on costs or guidance toward low-cost options. They often provide marginal product information, typically in Key Investor Information Documents that tend to create an illusion of information rather than offering substantial guidance (Oehler et al., 2023).

5. Discussion

ChatGPT provides investment recommendations that align with the consensus in the academic literature. As a robustness check, we conducted repeated queries to ChatGPT during our analysis to assess whether the recommendations remain consistent over time. We observed that the recommended asset allocations do change slightly over time (e.g., for investor profile C: 70 % in equity funds, 15 % in bond funds, and 15 % in commodities and REITs). It's worth noting that these changes do not necessarily imply that the asset allocation suggested by ChatGPT always results in the highest risk-adjusted returns. Temporary underperformance may be attributed to exogenous shocks in interest rates or stock prices, as well as differences in geographical distributions (Casta, 2023; Oehler et al., 2017; Mohammed et al., 2023). Nevertheless, based on empirical academic studies on portfolio optimization, the portfolios recommended by ChatGPT should not significantly underperform those of the robo-advisors over the long term (Jacobs et al., 2014; Horn and Oehler, 2020).

Our analysis of the three investor profiles indicates that ChatGPT's recommendations can guide retail investors toward earning the equity premium, whereas some robo-advisors may discourage retail investors from participating in the stock market. Non-participation in the stock market can be costly for retail investors (Mehra and Prescott, 1985; Siegel and Thaler, 1997), and ChatGPT can help mitigate the negative effects of non-participation for households. Importantly, ChatGPT recommends risky asset shares that align with investors' risk-bearing capacity, whereas some robo-advisors suggest higher risky asset shares. This can be problematic, as retail investors who have experienced significant financial losses tend to avoid risky assets in the future (Malmendier and Nagel, 2011; Malmendier, 2021). We also found that slight deviations in the answers to the questions of the robo-advisors do not significantly change the recommended asset allocation. Therefore, our findings regarding the usefulness of robo-advisors' recommendations can be considered robust.

An important finding for regulators, policymakers, and retail investors is that ChatGPT does not promote but rather discourages investing in high-fee products and costly rebalancing services. Both issues are financial pitfalls that can cost investors between 0.5 and 2.5 % of the risky asset share per annum (French, 2008; Horn and Oehler, 2020; Horn et al., 2020). Therefore, the net financial performance of the advice provided by ChatGPT should be higher, even when compared to the financial performance of robo-advisors before considering fees and rebalancing costs.

Our results align with those of Slater (2023), who concludes that it is possible to use ChatGPT effectively as a financial advisor by following certain rules and steps: "While it's important to remember that ChatGPT is not a professional financial advisor and cannot provide specific investment advice, it serves as a valuable resource for general information and guidance. With the help of ChatGPT, you can create a solid financial plan, make informed decisions, and work towards your financial goals with confidence."

However, there are also critical voices regarding ChatGPT as a financial advisor. Pino (2023) points out that ChatGPT has limited knowledge of events after 2021, does not account for changes in income, assumes that one knows how much one wants to invest, and has limitations in setting realistic goals. These concerns are valid points; however, some of them also apply to robo-advisors. Furthermore, all these concerns are only relevant for ongoing holistic financial management and not for one-time investments, which were the focus of our research question. Nevertheless, it would be interesting to see further research that compares the advice of ChatGPT and robo-advisors for more complex investment tasks, e.g., savings plans, tax loss harvesting etc.

6. Conclusion

Our analysis reveals the economic utility of generative AI-based techniques for analyzing and summarizing investment information for various investor profiles effectively. ChatGPT not only provides appropriate and well-founded portfolio recommendations for the three investor profiles but also delivers recommendations more quickly and with significantly less effort for the user. It is accompanied by easily understandable and valuable information about investing. An important policy implication is to clearly indicate that sophisticated independent chatbots can be recommended as a trustworthy source for at least a second opinion to double-check recommendations provided by a human or robo-advisor.

Our results also demonstrate that the business models of robo-advisors may face challenges if they do not offer more value to retail investors. One potential approach for robo-advisors to enhance their competitiveness is to integrate the advantages of ChatGPT or other chatbots with customer data to provide comprehensive financial advice and management. Robo-advisors could, for example, explore offering a rebalancing service throughout the customer's lifecycle by considering significant life events such as marriage, starting a family, home purchase, retirement, and more. This would involve automatically providing relevant information on financial products to customers *via* chatbots.

The findings of our study provide preliminary evidence that ChatGPT can significantly improve the presentation of investor information while maintaining and enhancing its informational content. In other words, ChatGPT enables investors to focus on the most relevant information. This suggests that generative AI systems have the potential to be the next significant advancement in financial advice and investor information. Therefore, our findings are not only relevant to academics but also to regulators, investors, and financial service providers. Future research could explore issues related to ESG investing and expand our research question to include human advisors and more complex investment decisions.

CRediT authorship contribution statement

Andreas Oehler: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Matthias Horn:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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